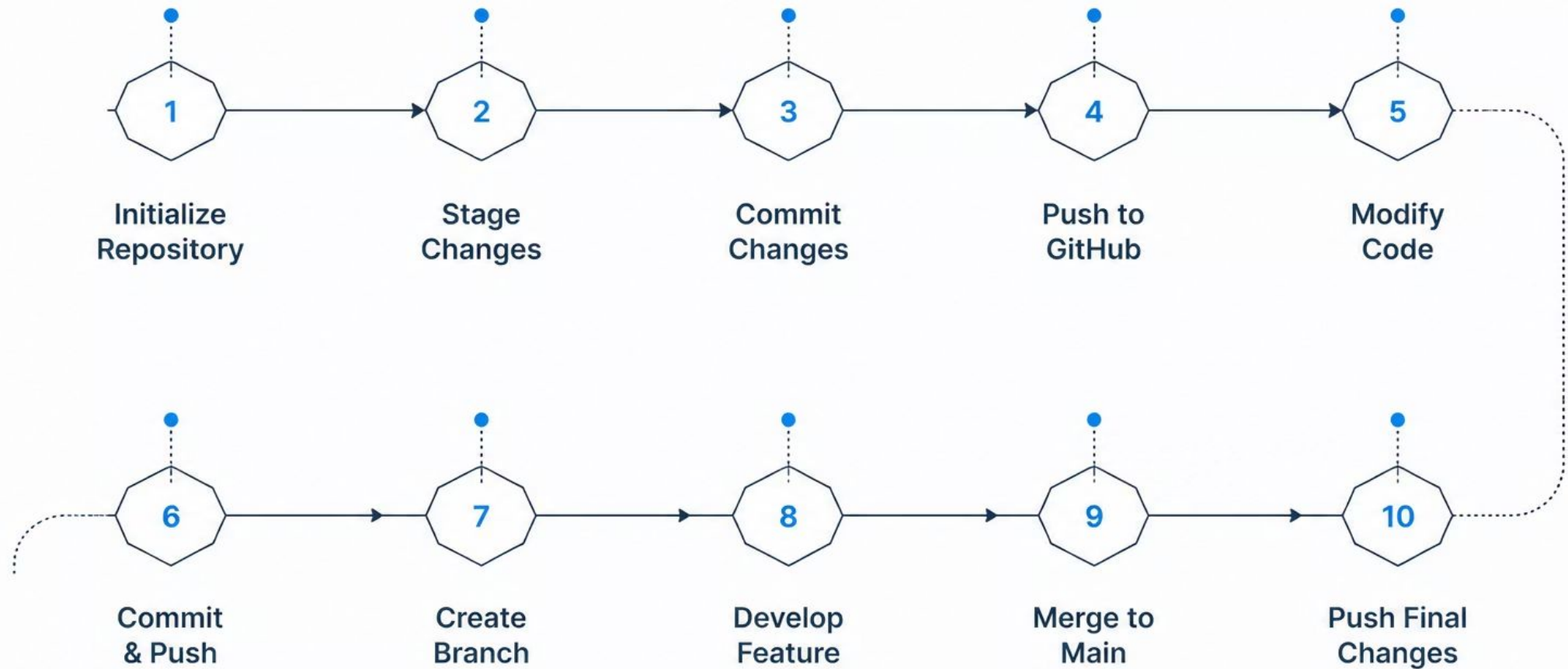


GIT WORKFLOW



GIT SETUP

Command	Description
<code>git init</code>	Initialize a new Git repository
<code>git config --global user.name "Your Name"</code>	Set your Git username
<code>git config --global user.email "you@example.com"</code>	Set your Git email

CHECK REPOSITORY STATUS

Command	Description
<code>git status</code>	Show the current status of the repository
<code>git branch</code>	Show all local branches
<code>git remote -v</code>	Show remote repository URLs

STAGING FILES

Command	Description
git add README.md	Stage a specific file
git add .	Stage all modified and new files
git restore --staged <file>	Remove a file from the staging area
git reset <file>	Unstage a file (alternative method)

COMMITTING CHANGES

Command	Description
git commit -m "First commit"	Create a commit with a message
git commit -m "Second commit"	Save another snapshot of your project

CONNECT TO GITHUB

Command	Description
<code>git remote add origin <repository-url></code>	Connect local repository to GitHub
<code>git remote -v</code>	Verify remote repository

Example: `git remote add origin https://github.com/username/repository.git`

PUSH TO GITHUB

Command	Description
<code>git push origin main</code>	Upload commits to GitHub

CLONING REPOSITORY

Command	Description
<code>git clone <repository-url></code>	Download an existing GitHub repository

Example: `git clone https://github.com/username/repository.git`

DISCARD CHANGES

Command	Description
<code>git restore README.md</code>	Restore a modified file to the last committed version

VIEW DIFFERENCES

Command	Description
<code>git diff</code>	Show unstaged changes
<code>git diff --staged</code>	Show staged changes before committing

BRANCHING

Command	Description
<code>git branch developer</code>	Create a new branch
<code>git branch</code>	List all branches
<code>git checkout developer</code>	Switch to another branch
<code>git branch -M main</code>	Rename current branch to main

MERGING

Command	Description
<code>git merge developer</code>	Merge the developer branch into the current branch

COMMIT HISTORY

Command	Description
<code>git log</code>	Show complete commit history
<code>git log -p -3</code>	Show last 3 commits with code changes

UPDATE LOCAL REPOSITORY

Command	Description
<code>git pull</code>	Fetch and merge changes from GitHub
<code>git fetch</code>	Download changes without merging

MERGE CONFLICT RESOLUTION

When a push is rejected because someone else has pushed changes:

`git pull`

If a conflict occurs:

- Open the conflicted file.
- Resolve the conflict manually.
- Save the file.

Run: `git add .`

`git commit -m "Resolved merge conflict"`

`git push origin main`